Group Project – Communications App

Software Requirements Specification

Revision History

Date	Revision	Description	Author
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1. Purpose

This document outlines the requirements for a Java Communications Client and Server Application.

1.1. Scope

This document will catalog the user, system, and hardware requirements for the Java Communications Client and Server Application system. It will not, however, document how these requirements will be implemented.

1.2. Definitions, Acronyms, Abbreviations

Communications App: A private text-messaging app that allows for asynchronous and synchronous communication manageable by IT.

TCP/IP: This stands for Transmission Control Protocol/Internet Protocol, which is a set of communication protocols used to connect devices on the internet and private networks.

Synchronous/Asynchronous: Chat history between users is persistent regardless if users are online or offline.

Help Page: The user manual for utilizing the Java GUI Client.

1.3. References

Use Case Specification Document – Step 2 in assignment description

UML Use Case Diagrams Document – Step 3 in assignment description

Class Diagrams – Step 5 in assignment description

Sequence Diagrams – Step 6 in assignment description

1.4. Overview

We are creating a Java Client and Server Communications Application that allows for synchronous and asynchronous text-based communication between several employees over TCP/IP, manageable by IT.

2. Overall Description

2.1. Product Perspective

Our multi-system communication application can serve a wide range of customers. Whether they be a small team, or as large as an enterprise, Harmony is there to ensure a seamless, fast, and secure experience. Being hosted in Java, a leading programming language that beats out C++ or Python in sheer resource management, competitors are simply out of the question.

2.2. Product Architecture

The system will be organized into 2 major modules: the **Client** module and the **Server** module. These modules may have sub-systems, some of which may be recognized later as potential class candidates later in design.

- Client Module:
    - User, user of the system.
    - IT, superuser of the system.
    - GUI, interactive user interface for clients.
    - Networking, to handle Inbound/Outbound traffic.
    - Help Page, a short page to explain the basic usages of the GUI.
    - ...
- Server Module:
    - ConsoleUI, to interact with any server-side processes.
    - Networking, to handle Inbound/Outbound traffic.
    - Logging, a record of all user chats.
    - Registration, logical processing of new users.
    - Verification, logical determination of existing users.
    - …

2.3. Product Functionality/Features

The high-level features of the system are as follows (see section 3 of this document for more detailed requirements that address these features):

Communication is strictly text-based. No images nor videos.

Communication is both asynchronous and synchronous.

Privacy between users and IT is negligible.

IT users must register any new users to the system.

Users need to have a way to log in and log out of the client.

Log ins must be verified, so that only registered users can use the client.

Help Page will cover the different client capabilities.

Client and Server are capable of two-way communication.

Users are allowed to converse with another user or several at once.

The message box supports conventional text modification, i.e. copy and paste, redo and undo.

Users will be able to search for other users to either individually chat or form a group with.

2.4. Constraints

Since users will directly interact with system by an application, the system must not contain browser specific code

Chat logs should be saved and stored by some means

Any new or additional users of the applications must be created by the IT

The application may only be programmed in Java.

The application's soft deadline is 04/30/2026.

The application's hard deadline is 05/05/2026.

The application cannot use databases.

The application cannot use encryption.

The application must use TCP/IP for network connections.

No code should be AI-generated or "vibe-coded".

Built-in data structures, unless otherwise stated or asked about beforehand, are not allowed.

Exceptions are: …

Developers may only use the Eclipse IDE for Java Developers.

Developers are expected to maintain standard coding conventions, including but not limited to: Good comments, "Clean code", Naming conventions, or Pre-conditions/Post-conditions for methods/function calls.

Users cannot have any access to logs.

2.5. Assumptions and Dependencies

It is assumed that the application will handle a few thousand users at any given time.

We can assume that the network when the application will be used is established on each user's device.

In addition, we must assume that the company will provide computers that are capable of running the communication applications, thus reducing the need for compatibility comparison.

The help page is assumed to not cover every last aspect of Client GUI.

It is assumed that the existing infrastructure within the company supports basic electronic devices.

It is assumed that since there is no encryption, the users of the system are completely perfect at keeping their credentials to themselves, and that there are never any cybercrimes or attacks committed against this software.

Both the server and client application will be dependent on the Eclipse IDE for Java Developers for their development.

3. Specific Requirements

3.1. Functional Requirements

3.1.1. Common Requirements:

3.1.1.1 The system must support TCP/IP based communication between both modules.
3.1.1.2 The system must ensure two-way connection between both modules, allowing both to send and receive data.
3.1.1.3 The system must gracefully handle network interruptions and notify users of disconnection.

3.1.1.4 The system must support and maintain both synchronous and asynchronous messaging.
3.1.1.5 The system must ensure that messages are properly stored and delivered to its intended recipient in chronological order.
3.1.1.6 The system must store all chat history on the server to be made accessible after the user disconnects or logs out.
3.1.1.7 The system must ensure that all logs and chat history is not accessible to non-IT users.

3.1.1.8 The system must maintain and enforce user authentication through a unique username and password before accessing messaging features.
3.1.1.9 The system must restrict access from unregistered users.
3.1.1.10 The system must log all login attempts, both successful and unsuccessful.

3.1.1.11 The system must allow multiple users to be connected simultaneously.
3.1.1.12 The system must associate each message with a sender identifier, a recipient identifier, and a timestamp.
3.1.1.13 The system must support both private messaging and group messaging between users.

3.1.1.14 The system must support and validate strictly text-based communication and reject any attempt otherwise.

1.1.1. Client Module Requirements:

Users must log in using a unique username and password, both of which are alphanumeric strings between 3 and 20 characters in length.

Must display a scrollable conversation history showing all sent and received messages with timestamps and sender labels for each message.

The client must provide a text input field and send a button for composing and transmitting text-based messages to selected recipients.

The client must display a contact's panel showing available users and existing chats, allowing users to select contacts for private messaging or group invitations.

Include a search feature allowing users to find other registered users by entering partial usernames, with matching results displayed for selection.

The client must provide an accessible Help Page from the main menu describing login procedures, sending messages, group chat management, and application restrictions.

The client must support conventional text editing features, including copy, paste, undo, and redo within the message area.

1.1.2. IT Module Requirements:

IT must provide an administrative interface accessible only after authentication with IT role privileges for user registration and history access.

IT must include a user registration form with fields for username, password, full name, and role selection for creating new user accounts.

IT must validate that passwords between 6-30 characters before submitting registration requests to the server.

IT must display success or error messages after each registration attempt, confirming new user records or reporting issues such as duplicate usernames.

The IT module must provide access to view any user's complete chat history upon request, displaying all messages with timestamps and participant information.

The IT module must display real-time server status information including active user connections and system health indicators.

1.1.3. Server Module Requirements:

The server must log all login attempts, with timestamps and usernames for security monitoring.

The server must process user registration requests from IT clients by creating new user records with unique usernames and storing them persistently in the user database.

The server must authenticate users by validating credentials against the stored user database before granting access to messaging features.

The server must process user registration requests from IT clients and create new user accounts with unique usernames.

The server must provide a console interface for server administrators to monitor active connections and system status.

The server must maintain group chat functionality, including group creation, member management, and message broadcasting to all group members.

1.2. External Interface Requirements

The system must support TCP/IP based communication to maintain authentication, messaging, and session management.

The Client must initiate a socket connection to the Server while the Server must continuously listen for incoming connection requests.

The system must maintain an active connection for the duration of each session and close upon logout. In the event of network failure or disconnection, the system must terminate the session, notify the Client, and detect the loss of disconnection.

The system must transmit authentication requests upon each user login before allowing access to messaging features.

All communication must follow a protocol consisting of a message identifier, a sender and recipient identifier, and timestamp. Messages must be encoded as plain text.

The system must provide a Java-based UI as the external interaction layer between the user and Client module.

The login interface must take input for a unique username and password. Upon submission, the interface must validate completion, verify authentication, and display a success or failure response. Messaging features are only accessible upon success.

The messaging interface must display the user's scrollable conversation history, a text input field for sending messages including a send action, and a messaging panel of users and existing chats.

The interface must provide an accessible Help Page that describes the functionality of the application, including login procedure, group chat behavior, messaging procedure, and general restrictions.

The system must provide an administrative interface for IT users to interact with Server-side processes. This must allow IT users to register new users with a unique username and password, creating and storing a new user record.

1.3. Internal Interface Requirements

The system must enforce immutable message storage by ensuring all file write operations use append mode only. No user, including IT, will have the ability to update or delete any stored messages.

The server must permanently store every text message by appending user specific history files. Each message record must include the timestamp, sender, recipient, and plain text messages. Storage will be done without any modification or deletion.

The server must maintain a text message queue for each offline user when offline. When users reconnect and log in, all queued text messages are delivered in the order they were received and remain permanently stored in the user's history

Only IT users have access to view text message history. The server must verify the role of the user's from the active session before granting access to the history file. Regular users requesting a history record will receive an access denied response

The server module must provide a shutdown that saves all offline text messages, closes all client connections, and releases all file handles before terminating

The Group management subsystem must maintain group data for each text-based group chat created in the system, including group name, admin username, and each member in the group username for each group chat created in the system

The server must reject any non-text based communication. Any messages attempting to include binary data, images, or media files must be rejected and the sender must recieve a error notification

The user must maintain user database that stores all registered usernames, password, and roles(admin, IT, etc)

The server must track and store the online/offline status of all registered users in real time. When connected or disconnected the server must update the sstatus

Server must maintain separate history files for private conversations and group conversations. Private chat history will store user specific files, while group char history must store group specific file containing all messages sent to the group with participatt information

4. Non-Functional Requirements

4.1. Security and Privacy Requirements

Example:
4.1.1 The System must encrypt data being transmitted over the Internet.

4.2. Environmental Requirements

Example:
4.2.1 System cannot require that any software other than a web browser be installed on user computers.
4.2.2 System must make use of the University's existing Oracle 9i implementation for its database.
4.2.3 System must be deployed on existing Linux-based server infrastructure.

4.3. Performance Requirements

Example:
4.3.1 System must render all UI pages in no more than 9 seconds for dynamic pages. Static pages (HTML-only) must be rendered in less than 3 seconds.